

Functional characterization of human inversions at the transcriptome and pangenome levels

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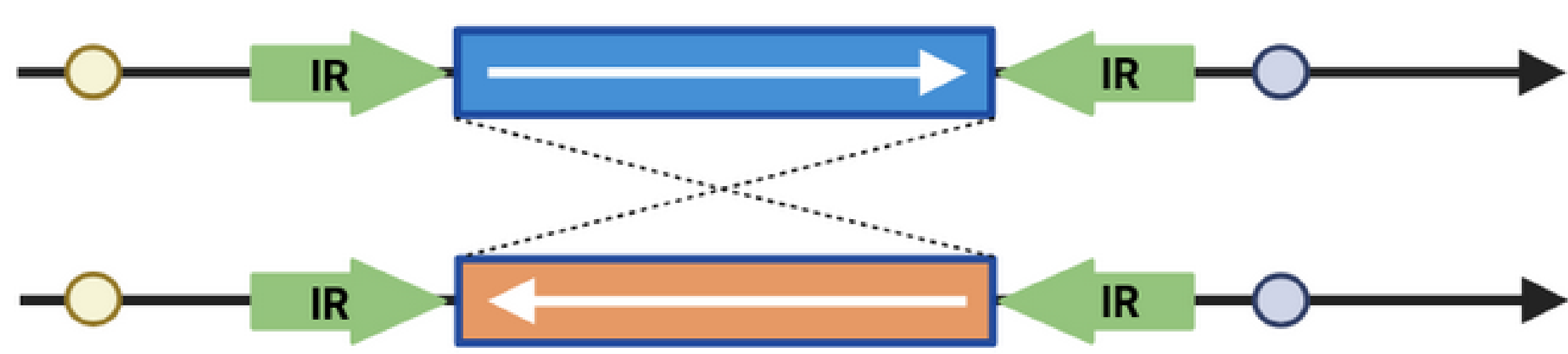
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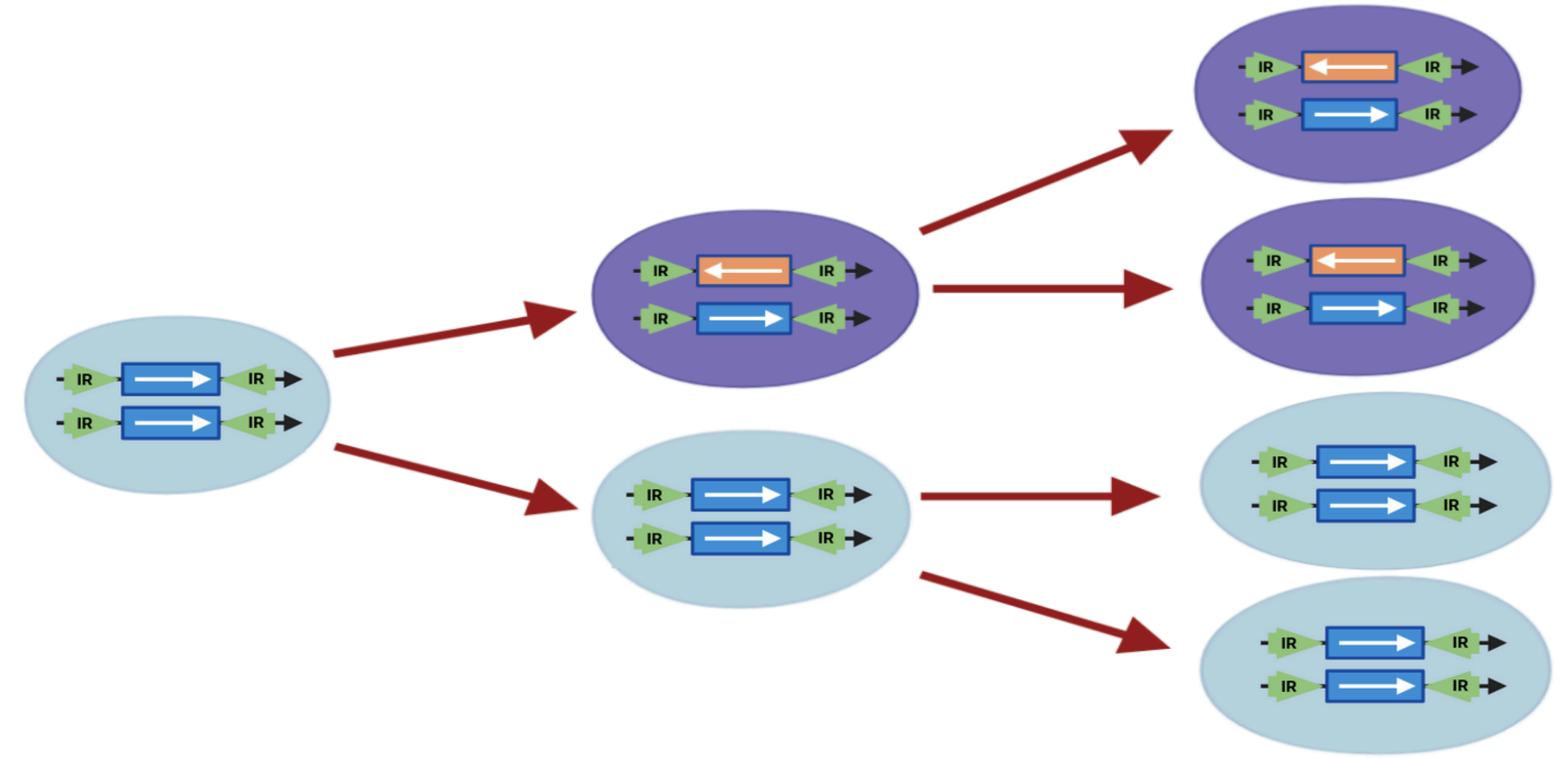
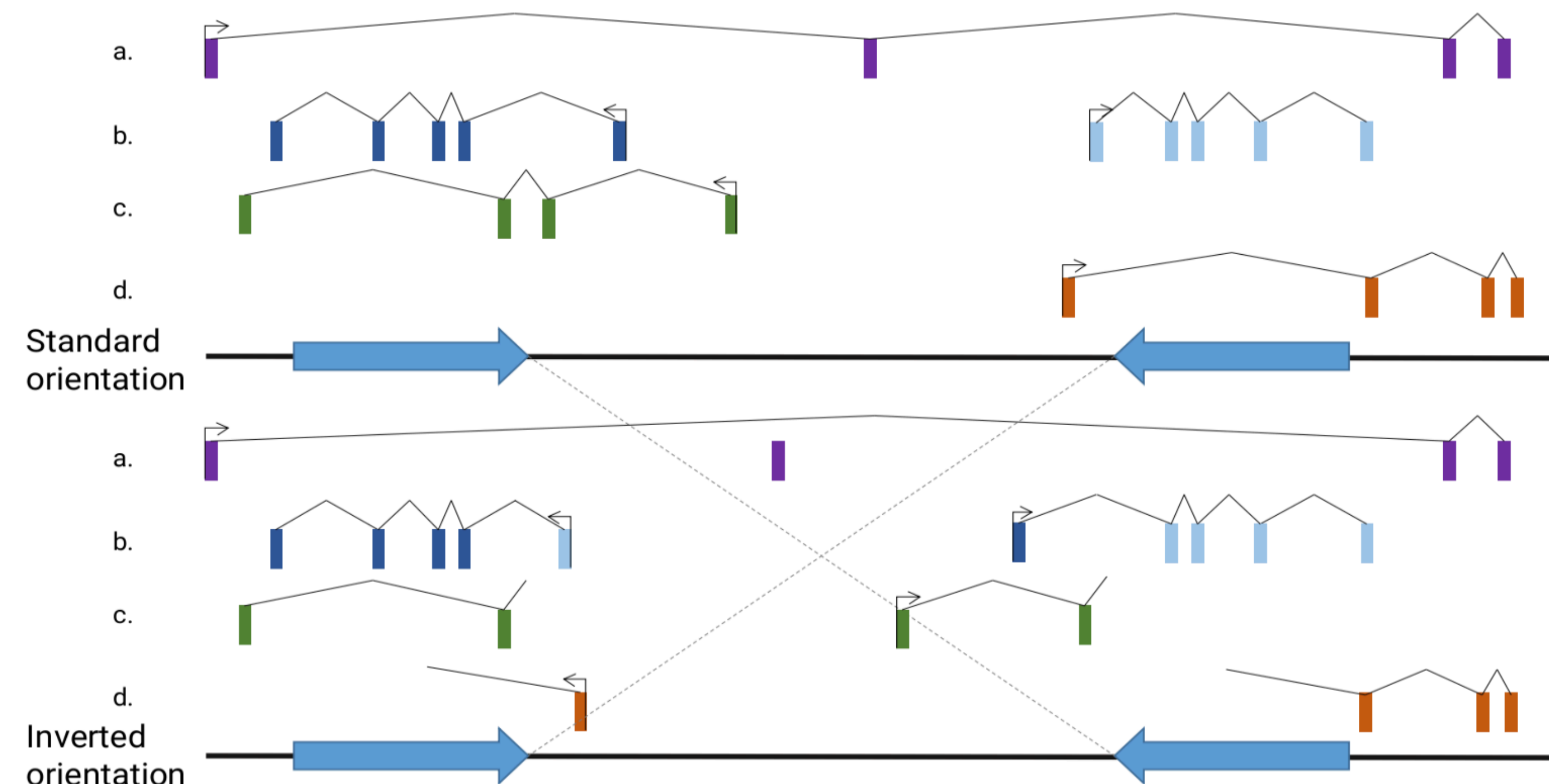
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Inversion transcriptomics and mosaicism



- Inversions flanked by inverted repeats (IRs) tend to be recurrent
- This recurrence can happen at a population level or at an individual level (mosaicism)
- The rate of inversion mosaicism and its functional implications is still unknown



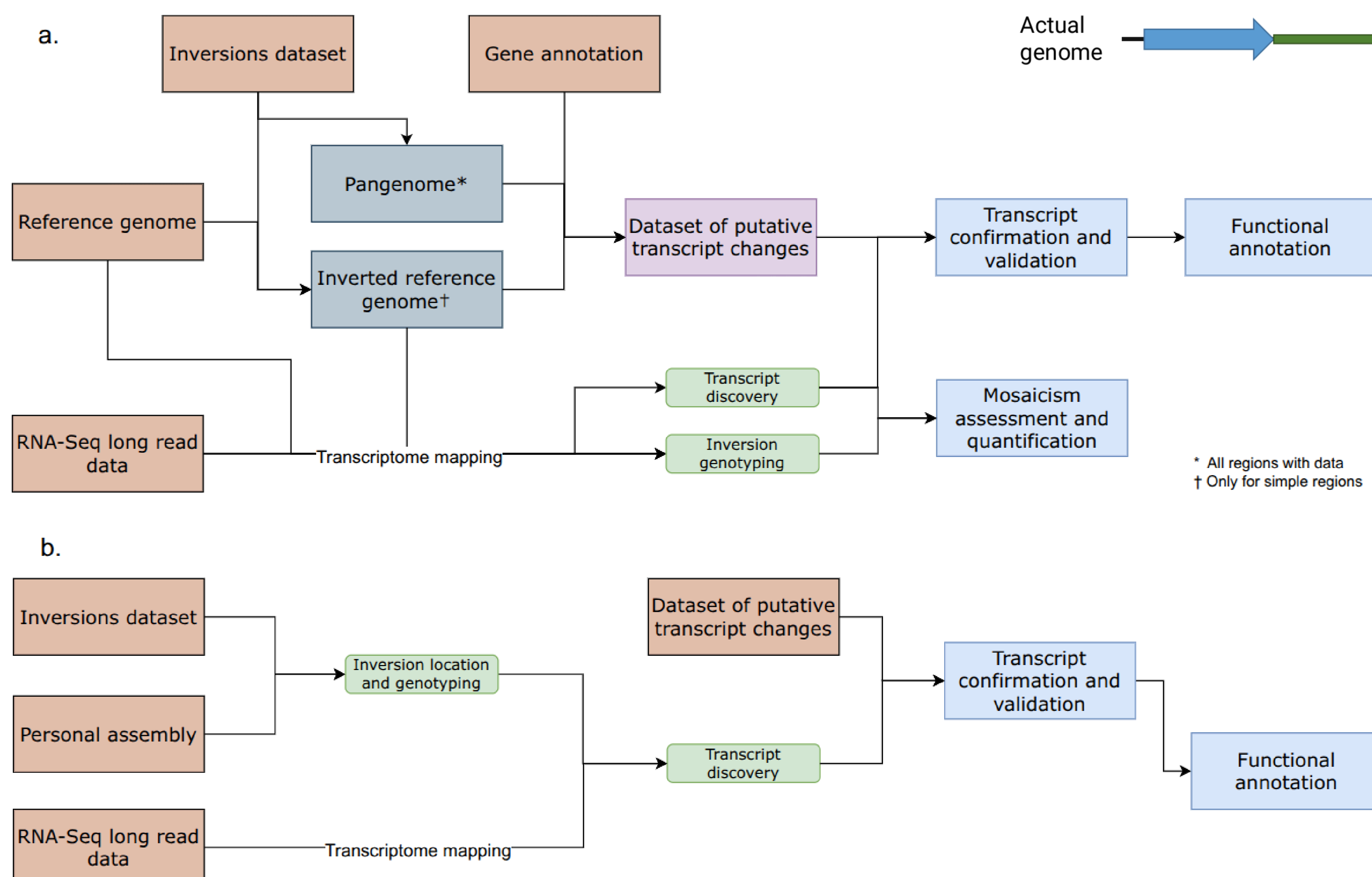
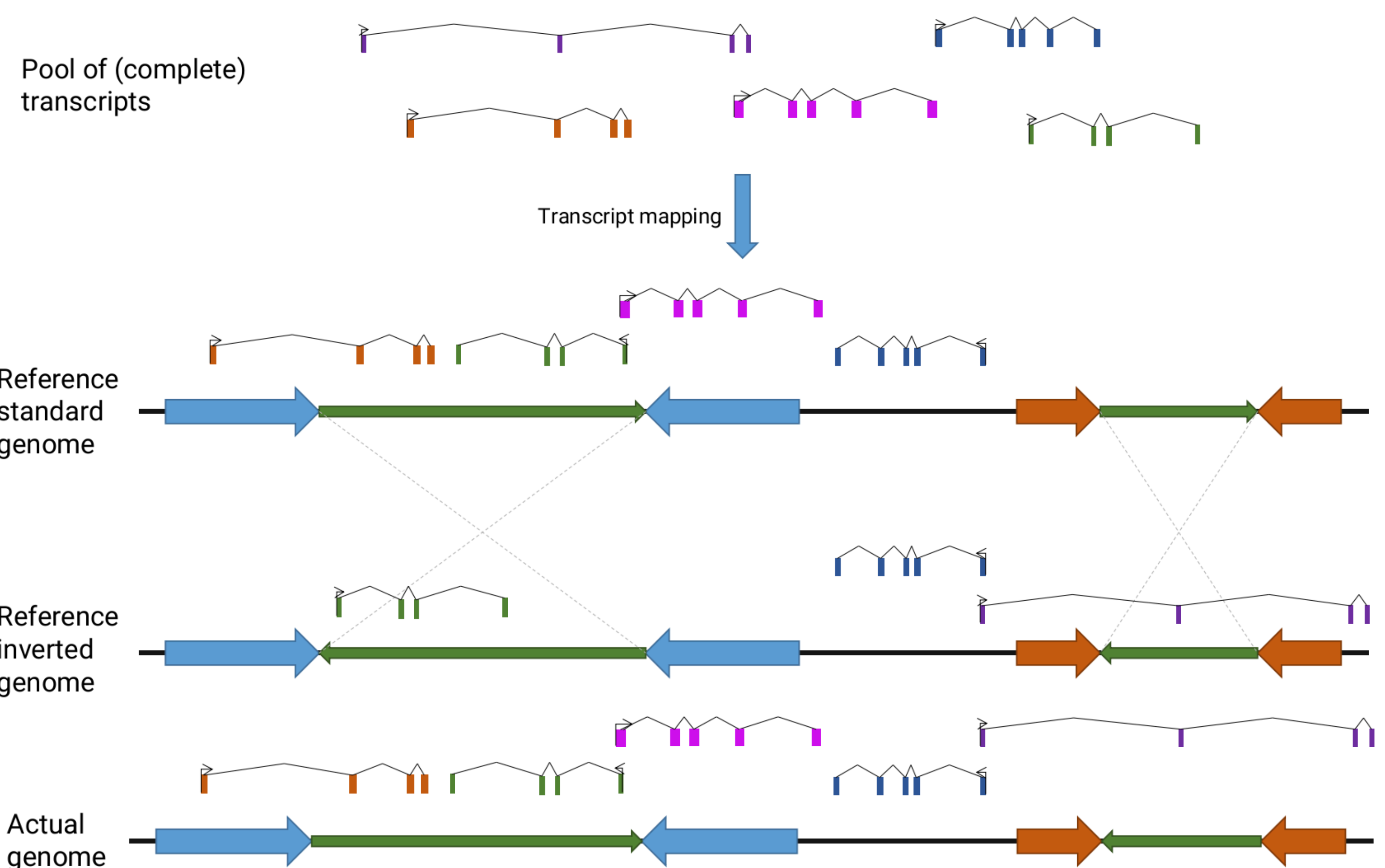
- Inversions disrupt the genomic landscape of the regions in which they occur
- The relative position of the gene with the inversion determines its possible effects on transcripts and exons, which include exon skipping (a), exon exchange (b) and transcript disruption (c, d)
- Mapping to a reference can miss new transcripts formed by inversions, and *de novo* annotation might be limited by the absence of these novel transcripts in current gene databases, hindering the determination of the effect of these transcripts

Methodology

- Using an inverted reference genome allows for the discovery and adequate mapping of novel transcripts that might not map or only map partially to the standard reference genome

- An individual genome is a combination of the standard and inverted reference genomes, therefore using standard and inverted reference genomes achieves a more comprehensive transcript mapping and discovery

- In complex regions with many haplotypes, the inverted reference genome is limited in its ability to capture some of the putative novel transcripts



- Whereas the addition of an inverted reference genome is feasible only for simple regions (a), the use of the pangenome (a) as a reference can be extended for complex regions, especially when the haplotype of the sample is contained within the pangenome

- The identification of orientation-specific transcripts enables inversion genotyping and, when combined with transcript quantification, allows estimation of mosaicism levels within a sample

- When available, the use of a personal assembly (b) as a reference offers the most accurate and comprehensive transcript mapping and discovery

- Combining inversion genotype information with transcript datasets generated from inversion-aware gene annotations, novel discovered transcripts can be validated

* All regions with data
† Only for simple regions